Project Synopsis

1. Title of Project

NexusFlow – Enterprise Project & Team Collaboration Management System

2. Problems with the Existing System

Existing enterprise project management solutions such as Jira, Trello, Asana, and ClickUp provide excellent project management capabilities but are often expensive, overly complex for small organizations, or require multiple third-party integrations. NexusFlow aims to provide an integrated, lightweight, and role-based collaboration platform suitable for startups, educational institutions, and small-to-medium enterprises.

This fragmented approach leads to several practical problems:

1. Data Inconsistency: Project information is scattered across multiple platforms, making it difficult to maintain a single source of truth.

2. Lack of Accountability: Without a centralized task assignment system, it becomes difficult to track who is responsible for which task and by when it should be completed.

3. Poor Visibility: Project managers and stakeholders lack a real-time, unified view of project progress, task status, and team workload.

4. Communication Gaps: Important discussions related to specific tasks get lost in general chat applications, disconnected from the actual work item they relate to.

5. Manual Reporting: Generating progress reports requires manually compiling data from multiple sources, which is time-consuming and error-prone.

6. No Role-Based Access: Existing manual systems do not enforce role-based permissions, making it difficult to control who can view, edit, or approve specific project information.

These limitations reduce productivity, increase the chances of missed deadlines, and make it difficult to scale project management practices as a team grows.

Why This Topic is Chosen (Present State of the Art)

The increasing adoption of digital collaboration platforms has transformed the way organizations manage projects and teams. Modern businesses require centralized systems that support task management, secure communication, project tracking, and real-time collaboration. Although several commercial solutions such as Jira, Trello, Asana, and ClickUp are available, they are often expensive, feature-heavy for small organizations, or require multiple third-party integrations. This project has been selected to develop a lightweight, secure, and integrated web-based collaboration platform that demonstrates modern software engineering practices while addressing the needs of startups, educational institutions, and small-to-medium enterprises.

3. Description of the Proposed System

- NexusFlow is a web-based, role-driven Enterprise Project and Team Collaboration Management System designed to unify project management, task tracking, team collaboration, and reporting into a single integrated platform.

- The system allows organizations to create projects, assign tasks to team members, track task progress through defined statuses, collaborate through task-specific comments, and receive real-time notifications for important updates such as task assignments.
- **NexusFlow follows a three-tier client-server architecture:** Angular (Frontend) → REST API (ASP.NET Core) → SQL Server (Database)
- The backend is built using ASP.NET Core (.NET 10) following Clean Architecture principles, with clear separation between the Domain, Application, Infrastructure, and API layers. This ensures the system is maintainable, testable, and scalable for future enhancements.
- The frontend is built using Angular, providing a responsive, single-page application experience for end users across devices.
- Authentication and authorization are handled through JSON Web Tokens (JWT), ensuring secure, stateless access control based on user roles.

Objectives of the Project

1. To develop a centralized web-based platform for managing projects, tasks, and team collaboration.
2. To provide secure authentication and authorization using JSON Web Tokens (JWT).
3. To enable project managers to assign, monitor, and track project tasks efficiently.
4. To improve communication among team members through task-specific comments and notifications.
5. To provide real-time project progress monitoring using dashboards and reporting features.
6. To reduce manual effort involved in project management and reporting.
7. To develop a scalable and maintainable application using Clean Architecture principles.

Scope of the Project

NexusFlow is designed to provide an integrated platform for enterprise project and team collaboration. The scope of the project includes user authentication, role-based authorization, project management, task management, task comments, notifications, dashboard reporting, and organization management. The system is intended for small-to-medium organizations, educational institutions, and startups requiring a centralized project management solution. The initial version focuses on web-based access using Angular, ASP.NET Core, and Microsoft SQL Server, while allowing future expansion through cloud deployment and mobile application support.

4. Description and Identification of the Functional Modules

NexusFlow consists of the following core functional modules:

1. Authentication & Authorization Module Users: All Users Allows users to register, log in, and securely access the system using JWT-based authentication. Supports role-based access control for Admin, Project Manager, Team Member, and Client roles.

2. User Management Module Users: Admin Allows administrators to manage user accounts, view user profiles, and control account status.

3. Organization Management Module Users: Admin Allows creation and management of organizations that group projects and users under a single entity.

4. Project Management Module Users: Admin, Project Manager Allows creation, updating, and tracking of projects including project status, timelines, and assigned members.

5. Task Management Module Users: Project Manager, Team Member Allows creation and assignment of tasks within a project, including priority levels, due dates, and status tracking (To Do, In Progress, In Review, Done).

6. Comment & Collaboration Module Users: Project Manager, Team Member Allows team members to discuss specific tasks through task-linked comments, keeping all communication contextual to the relevant work item.

7. Notification Module Users: All Users Automatically notifies users of important events such as task assignments, ensuring no critical update is missed.

8. Dashboard & Reporting Module Users: Admin, Project Manager, Client Provides a consolidated view of project statistics, task progress, and team performance.

5. Tools / Platforms

5.1 Hardware Specifications:

Processor: Intel Core i5 / AMD Ryzen 5 or higher

RAM: 8 GB minimum (16 GB recommended)

Storage: 100 GB free disk space (SSD preferred)

Display: 1366 x 768 resolution or higher

Operating System: Windows 10/11 (64-bit)

5.2 Software Specifications:

Frontend: Angular, TypeScript, HTML5, CSS3, Bootstrap, Angular Material

Backend: ASP.NET Core (.NET 10), C#, Entity Framework Core, AutoMapper, FluentValidation

Database: Microsoft SQL Server 2022 (Developer Edition)

Authentication: JSON Web Tokens (JWT)

Development Tools: Visual Studio 2022, Visual Studio Code, SQL Server Management Studio (SSMS)

Version Control: Git, GitHub

API Testing: Postman

Architecture: Clean Architecture (Domain, Application, Infrastructure, API layers)

Design Pattern: Repository Pattern, Unit of Work Pattern

6. Methodology

6.1 SDLC Model to be Used:

The Incremental Model has been selected for the development of NexusFlow.

Development and Testing Methodology

The development of NexusFlow follows a structured software development process consisting of Requirement Analysis, System Design, Development, Testing, and Deployment. During the analysis phase, project requirements and user roles were identified. The design phase focuses on Clean Architecture, database schema design, and RESTful API planning. Development is carried out incrementally, allowing each module to be completed and verified independently. Testing is performed after each increment using Postman for API testing, SQL Server Management Studio (SSMS) for database verification, and browser-based testing for frontend validation. This approach helps identify defects early and ensures the reliability of each completed module.

6.2 Justification for the Selection of Model:

NexusFlow consists of multiple independent but interconnected modules such as Authentication, Project Management, Task Management, Comments, and Notifications. The Incremental Model allows each module to be designed, developed, and tested as a self-contained increment before moving to the next, which offers several advantages for this project:

1. Since the project is being developed individually, the Incremental Model allows focused development on one module at a time, reducing complexity and the risk of errors.

2. Each completed increment (such as the Authentication module) results in a working, testable piece of functionality, allowing early verification through tools like Postman before proceeding further.

3. Given the fixed academic submission deadlines, the Incremental Model ensures that even if time constraints arise, the core modules completed earlier remain fully functional and demonstrable.

4. This model aligns naturally with the project presentation structure required by the institute, where an initial presentation is given after the design phase and a second presentation after system testing is complete.

For these reasons, the Incremental Model was determined to be the most suitable approach for developing NexusFlow within the available timeframe.

Contribution / Value Addition

NexusFlow provides a centralized enterprise collaboration platform that combines project management, task tracking, team communication, and reporting within a single application. The system reduces dependency on multiple software solutions, improves team accountability through role-based task assignments, enhances communication using task-specific comments, and simplifies project monitoring through dashboards and notifications. The project also demonstrates the practical implementation of modern software engineering concepts including Clean Architecture, Repository Pattern, Unit of Work Pattern, JWT Authentication, and RESTful Web APIs.

Limitations of the Project

The current version of NexusFlow is limited to web-based access and requires an active internet connection for operation. Real-time collaboration features such as live notifications using SignalR are not included in the initial release. Integration with third-party services such as GitHub, Microsoft Teams, and cloud storage platforms is planned for future versions. Mobile application support is also outside the scope of the current implementation.

7. Conclusion & Future Scope

Conclusion

NexusFlow addresses a genuine gap in how small organizations and student teams manage projects — the fragmentation of task tracking, communication, and reporting across disconnected tools. By implementing Clean Architecture on ASP.NET Core (.NET 10), role-based JWT authentication, and an Angular front end backed by SQL Server, the project demonstrates the Software Engineering discipline (SRS, DFD, ER modelling, incremental development, structured testing) required by the training guidelines alongside a technically sound, industry-relevant full-stack implementation. The incremental approach ensures a working, demonstrable system exists at every stage of the six-week training period.

Future Scope

While the current scope of NexusFlow covers the core modules necessary for project and team collaboration, the system has been designed with extensibility in mind. Potential future enhancements include:

1. Real-time notifications and live updates using SignalR for instantaneous communication between users.

2. File attachment support for tasks and projects using cloud storage integration such as Azure Blob Storage.

3. Calendar and meeting scheduling module for coordinating team availability and deadlines.

4. Advanced analytics and reporting with visual dashboards using Chart.js, including burn-down charts and team performance metrics.

5. Mobile application support through a Progressive Web App (PWA) implementation of the Angular frontend.

6. AI-based task prioritization that suggests optimal task ordering based on deadlines, dependencies, and team workload.

7. Integration with third-party tools such as GitHub for linking code commits directly to tasks.

8. References

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